A Joint Masked Spectral Group Framework for Hyperspectral Representation Learning

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***Abstract*—Given the scarcity of labeled data, hyperspectral image (HSI) analysis has increasingly relied on self-supervised representation learning. Masked reconstruction and contrastive learning are two dominant paradigms that emphasize spectral–spatial fidelity and feature discrimination, respectively, yet their objectives are not naturally aligned in HSI: reconstruction is vulnerable to redundancy-induced shortcut solutions, while contrastive learning may under-emphasize subtle but class-critical spectral cues. Motivated by this gap, we propose a joint masked spectral group (MSG) framework for hyperspectral representation learning that couples the two objectives through a shared spectral-group representation and objective-consistent structured masking. At its core, MSG builds on spectral group embedding and masking (SGEM), which organizes contiguous bands into compact groups and performs group-wise masking to suppress redundancy while preserving coherent spectral context for stable view formation. This design encourages reconstruction and discrimination to complement rather than interfere with each other, yielding representations that are both discriminative and transferable. Extensive experiments under linear evaluation and few-shot fine-tuning (FT) across multiple downstream benchmarks demonstrate that MSG consistently outperforms strong masked reconstruction and contrastive learning baselines, achieving higher accuracy, more stable class-wise performance, and stronger transferability.**

***Index Terms*—Contrastive learning, hyperspectral image (HSI), masked reconstruction, self-supervised learning (SSL), spectral grouping, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide continuous spectral responses with fine-grained material information, supporting diverse applications such as land-cover classification, vegetation monitoring, environmental assessment, and urban or disaster surveillance [1], [2], [3]. To fully exploit this rich spectral–spatial information, however, several key challenges must be addressed. The high dimensionality of hundreds of correlated bands, variations across sensors and acquisition conditions, and the presence of noise and mixed pixels complicate reliable data interpretation [4], [5]. Moreover, the scarcity of large-scale, high-quality labeled data, together with the heterogeneity of acquisition conditions, further limits the scalability and robustness of hyperspectral analysis in practice [6], [7].

Over the past decades, HSI analysis has evolved through several methodological stages. Early studies relied on handcrafted features and shallow classifiers, including spectral indices [8], subspace-based methods [9], dictionary learning [10], and kernel-based support vector machines [11]. With the advent of deep learning, convolutional neural networks (CNNs) and their 2-D/3-D extensions became dominant, effectively capturing spectral–spatial correlations and achieving substantial gains in classification accuracy [12], [13]. More recently, attention-based architectures and transformer variants have further advanced HSI representation learning by enabling the modeling of long-range dependencies across spectral and spatial dimensions [14], [15]. Despite these advances, the strong reliance on annotated data remains a critical bottleneck, motivating the exploration of unsupervised and self-supervised paradigms for more generalizable hyperspectral representation learning.

Self-supervised learning (SSL) has emerged as a promising solution for HSI analysis, as it enables the extraction of transferable representations from abundant unlabeled data [16], [17]. In computer vision (CV), SSL has evolved into several dominant paradigms, such as masked reconstruction [18], contrastive learning [19], clustering-based representation learning [20], and generative pretraining [21]. When transferred to HSI, these paradigms have achieved encouraging progress but still struggle to fully capture the hierarchical spectral–spatial dependencies inherent in hyperspectral data [22], [23]. This limitation largely stems from the predominant reliance on a single pretext objective, which restricts the ability to exploit complementary learning cues. Specifically, masked reconstruction focuses on fine-grained spectral–spatial reconstruction, whereas contrastive learning emphasizes global feature consistency between augmented views. While a few studies have explored multiobjective SSL frameworks, they typically combine different objectives only at the loss level, leaving the two objectives to compete on inconsistent inputs and learning signals, which often yields limited synergy [24]. These limitations highlight the need for an SSL

Received 5 February 2026; accepted 10 March 2026. Date of publication 17 March 2026; date of current version 3 April 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFC3806000 and Grant 2023YFC3806002, in part by the National Natural Science Foundation of China under Grant 62371342 and Grant 62406227, in part by Shanghai Pujiang Program under Grant 25PJD128, and in part by the Fundamental Research Funds for the Central Universities. *(Corresponding author: Hao Deng.)*

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Data is available on-line at https://github.com/Viento1027/MSG
Digital Object Identifier 10.1109/TGRS.2026.3674733

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framework that can explicitly coordinate reconstruction- and contrastive-driven supervision in a shared spectral–spatial representation space, rather than merely summing losses.

To address these limitations, we propose the joint masked spectral group (MSG) framework for self-supervised hyperspectral representation learning. Instead of treating masked reconstruction and contrastive learning as two loosely coupled branches, MSG introduces an objective-consistent coordination mechanism that aligns their supervision through a shared spectral-group representation and structured masking. Concretely, MSG uses a shallow shared encoder to establish a common spectral–spatial feature space, upon which two lightweight task heads are optimized with different masking ratios to provide complementary learning signals. At its core lies the spectral group embedding and masking (SGEM) module, which aggregates contiguous spectral bands into coherent group-wise tokens and applies group-wise masking aligned with the grouped representation. In this way, reconstruction emphasizes intragroup spectral–spatial fidelity while contrastive learning promotes global invariance on consistent grouped views, so that the two objectives complement rather than interfere with each other.

The main contributions of this work are summarized as follows.

1) We propose MSG, a joint self-supervised framework for hyperspectral representation learning. It integrates spectral-group masked reconstruction and group-wise contrastive learning within a partially shared encoder architecture to balance fidelity and discriminability.
2) We design the SGEM module, which employs a spectral stem and group-wise patchification to partition contiguous spectral bands into groups, generating coherent spectral–spatial tokens. This module allows the transformer to jointly capture spatial and spectral correlations and provides a unified foundation for both reconstruction and contrastive learning.
3) We comprehensively evaluate MSG across multiple benchmark datasets through comparative and ablation analyses. The results demonstrate its consistent superiority over strong baselines and verify the generalization and robustness of the proposed framework.

The remainder of this article is organized as follows. Section II provides a review of related work. Section III describes the proposed MSG methodology. Section IV presents the datasets and experimental setup, and then reports the performance comparison and ablation studies. Section V concludes the article and discusses potential future research directions.

II. Related Work

A. Masked Reconstruction for HSIs

Masked reconstruction has become a prominent paradigm in self-supervised representation learning, which aims to recover missing input regions from partial observations [25]. In CV, this principle has been instantiated in various architectures, among which masked autoencoders (MAEs) are particularly influential [26]. Key design principles include employing a high masking ratio to increase task difficulty, adopting an asymmetric encoder–decoder structure with most capacity allocated to the encoder, and performing reconstruction either in the pixel domain or in compact latent spaces [27], [28]. These advances demonstrate that such reconstruction objectives can effectively capture contextual dependencies without reliance on manual annotations.

When extended to an HSI, masked reconstruction needs to account for the intrinsic spectral–spatial structure. Prior works have attempted to reconstruct full spectral signatures [22], low-dimensional subspace projections [29], or hybrid targets combining spectral and spatial cues [30]. Spatial masking encourages the modeling of local context, whereas spectral masking compels the encoder to capture interband correlations. Several studies further integrate spatial and spectral masking to model joint dependencies [31]. Decoders are generally designed to be lightweight, thereby avoiding trivial copying and forcing the encoder to assume the representational burden [32]. Nevertheless, recent analyses indicate that reconstruction with pixel-space or other low-level targets does not necessarily induce semantically meaningful high-level representations, even under high masking ratios [33]. Consistent observations further show that MAE-style pretraining often produces features with limited abstraction and entanglement of object and background cues, thereby increasing reliance on labeled fine-tuning (FT) for achieving strong downstream performance [34]. It also exhibits limited cross-sensor transferability, as fidelity objectives alone do not ensure domain invariance [35]. Consequently, reconstruction-based pretraining remains constrained in semantic abstraction and cross-domain generalization in HSI.

B. Contrastive Learning for HSIs

In parallel, contrastive learning has emerged as a central paradigm for self-supervised representation learning. By maximizing agreement between positive pairs and enforcing separation from negatives, it produces compact, discriminative, and transferable embeddings [36]. Technical innovations such as momentum encoders, memory queues, and temperature scaling have substantially improved its scalability and effectiveness in large-scale visual representation learning [19], [37].

In the hyperspectral domain, contrastive learning presents both opportunities and unique challenges. Spatial augmentations (e.g., cropping, flipping, and translation) enrich spatial diversity, whereas spectral perturbations (e.g., band dropout, jittering, and smooth distortions) expand the view space while preserving spectral structure [38]. The design of augmentations is particularly delicate: they must increase variability while maintaining the physical continuity of spectral signatures. To improve scalability and stability, momentum-based encoders and prototype-based clustering have been adapted to hyperspectral settings [39], [40]. Despite these efforts, several issues persist. Highly similar spectra across classes can generate false negatives, undermining the contrastive objective. Excessive spectral perturbations may destroy essential class-discriminative cues, while insufficient perturbations fail to provide meaningful contrast. Moreover, large-batch training and reliance on memory banks introduce computational burdens for high-resolution hyperspectral datasets [41]. These

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limitations highlight that contrastive learning for HSI remains sensitive to augmentation design, spectral redundancy, and computational scalability, calling for further exploration of more adaptive and domain-aware strategies.

C. Spectral Information Handling in Hyperspectral Representation Learning

A fundamental challenge in hyperspectral representation learning lies in efficiently encoding spectral information while capturing spectral correlation and preserving spatial structure. Early methods treated each band as an independent channel, which resulted in sequence lengths proportional to the number of bands or required computationally demanding 3-D convolutional networks [42], [43]. While effective in modeling fine-grained dependencies, these approaches are computationally expensive and prone to overfitting and noise.

Later studies introduced spatial-only patchification strategies inspired by vision transformers [44]. Although this improves computational efficiency, it weakens spectral correlation and undermines the advantages of hyperspectral data in complex spectral scenarios. To address this, more targeted spectral modeling techniques—such as channel attention, band selection, clustering, and subspace methods—have been explored to reduce redundancy and enhance robustness [45], [46], [47]. However, these strategies inevitably involve tradeoffs among fidelity, efficiency, and generalization.

In self-supervised pretraining, spectral handling becomes even more critical, because redundancy across bands can lead to shortcut solutions in masked reconstruction. Existing masking strategies typically rely on random band masking or contiguous block masking [48], [49]. Random masking may disrupt spectral continuity and leave highly similar bands unmasked, enabling the model to reconstruct masked bands by referencing redundant spectra. To mitigate such redundancy-induced information leakage, mining redundant spectra (MRS) introduces a redundancy-aware spectral masking policy that jointly masks highly similar bands selected via band-wise similarity, thereby increasing reconstruction difficulty and serving as a plug-and-play replacement for spectral-wise masking [50]. These redundancy-aware designs primarily refine the masking policy in masked reconstruction by determining which raw spectral components should be jointly hidden. Blockwise spectral embedding has also been explored to improve spectral continuity in masked pretraining [51], but it directly partitions raw bands without explicitly enhancing interband representation and remains primarily reconstruction-oriented.

Overall, effective spectral representation in hyperspectral pretraining requires not only preserving local continuity but also capturing higher level spectral relationships and cross-region coherence, while aligning spectral and spatial dependencies. Achieving such a balance remains challenging within existing designs, motivating the exploration of structured spectral modeling schemes that can organize correlated bands into coherent representations and better align spectral and spatial dependencies. Based on these insights, we propose a dual-task pretraining framework that coordinates masked reconstruction and contrastive learning via objective-consistent spectral-group representations and structured masking, aiming to obtain balanced spectral–spatial representations for hyperspectral pretraining.

III. Methodology

A. Overall Framework

To learn transferable representations from unlabeled HSIs, we present the MSG framework, which couples masked reconstruction and group-level contrastive learning through objective-consistent spectral-group representations and structured masking.

Raw hyperspectral patches $\tilde{\mathbf{X}} \in \mathbb{R}^{H_0 \times W_0 \times C_0}$ are first processed by group-wise principal component analysis (GWPCA) followed by spatial sampling. This step reduces spectral redundancy and yields preprocessed inputs $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ for the subsequent SGEM module.

As illustrated in Fig. 1, the SGEM module partitions contiguous spectral bands into groups and converts X into structured group-wise tokens, preserving intragroup spectral continuity while encoding local spatial structure. This design aggregates contiguous bands to enrich token information content, while avoiding the excessive sequence length of band-wise tokenization and maintaining computational efficiency.

The grouped tokens are then fed into a shallow shared encoder to learn task-agnostic spectral–spatial representations. These representations serve as a common space for two complementary branches: one focusing on fine-grained reconstruction and the other on global contrastive discrimination. The reconstruction branch adopts a relatively high mask ratio τ_h and employs an asymmetric decoder to recover masked group-patch values, enforcing fine-grained spectral–spatial fidelity. In contrast, the contrastive branch applies a lower mask ratio τ_l together with mild stochastic perturbations to generate correlated group-level views for contrastive optimization, thereby enhancing feature discriminability and robustness. During pretraining, these two objectives are jointly optimized through a weighted loss.

After pretraining, both the decoder and the projector are discarded. For downstream applications, encoder outputs are aggregated at the group level, such as by mean pooling or lightweight feature aggregation and forwarded to a task-specific prediction head.

B. Spectral Group Embedding and Masking

While the preprocessing step reduces redundancy, directly treating each spectral band as an independent token still leads to excessive sequence length, whereas purely spatial patchification neglects spectral correlations. To address these limitations, we design an SGEM module that first performs channel expansion and group-wise patchification to form coherent spectral–spatial tokens, and then applies structured masking aligned with the grouped representation. This process is illustrated in the lower part of Fig. 1, which depicts the transformation from the input hyperspectral sample to grouped tokens and their masked variants.

1) Spectral Stem: Let $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ denote a local hyperspectral sample. Before spectral grouping, a lightweight spectral stem ϕ_s composed of a 1×1 convolution, batch

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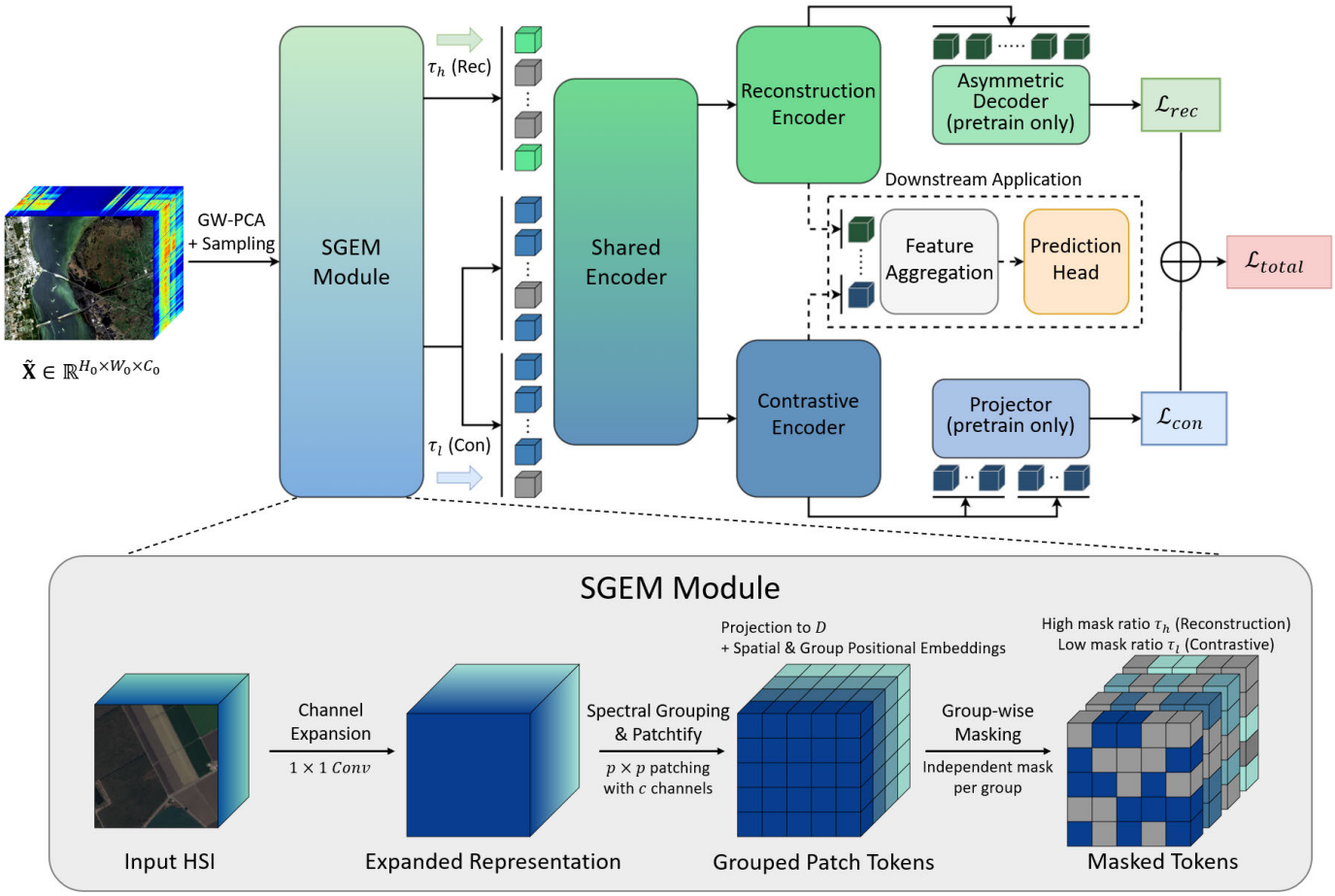


Fig. 1. Overall architecture of the proposed MSG framework. The model performs coordinated dual-objective pretraining, comprising a reconstruction objective with a high masking ratio and a contrastive objective with a low masking ratio. The SGEM module converts hyperspectral samples into spectral-grouped patch tokens, which are fed into the shared encoder and branch-specific heads. The bottom illustrates the detailed workflow of the SGEM module.

normalization, and GELU activation is applied along the spectral dimension

$$\mathbf{X}' = \phi_s(\mathbf{X}) \in \mathbb{R}^{H \times W \times C'}, \quad C' \geq C. \tag{1}$$

This operation performs pointwise channel mixing to linearly project each spectral vector into a more expressive subspace, while normalization and nonlinear activation further stabilize training and capture subtle interband dependencies. Serving as a spectral preprocessing stage, the stem enhances spectral decorrelation and feature discriminability before grouping, without introducing additional spatial computation. Its effect is more evident in improving feature adaptation and optimization stability, as it facilitates the alignment of spectral representations across varying domains and training dynamics.

2) Group-Wise Patchification: The spectral dimension of $\mathbf{X}'$ is partitioned into G groups, each containing $C_g = C'/G$ contiguous bands by zero-padding if needed. The spatial domain is divided into nonoverlapping patches of size $p \times p$, yielding $T = (H/p)(W/p)$ patches per group. Stacking across groups gives $N = G \cdot T$ tokens in total.

For each group g, the corresponding patch matrix is

$$\mathcal{P}_g(\mathbf{X}') = \left[\, \mathrm{vec}\left(\mathbf{X}'_{g,1}\right), \ldots, \mathrm{vec}\left(\mathbf{X}'_{g,T}\right) \,\right]^\top \in \mathbb{R}^{T \times p^2 C_g} \tag{2}$$

and concatenating all groups yields

$$\mathcal{P}(\mathbf{X}') = \begin{bmatrix} \mathcal{P}_1(\mathbf{X}') \\ \vdots \\ \mathcal{P}_G(\mathbf{X}') \end{bmatrix} \in \mathbb{R}^{N \times p^2 C_g}. \tag{3}$$

This operation preserves both spatial ordering and spectral grouping, ensuring that tokens encode contiguous spectral information while remaining computationally tractable.

3) Group-Specific Token Projection and Positional Embeddings: Instead of using a single shared projection, group-specific projections are adopted (equivalent to a grouped $p \times p$ convolution with stride p). For each group g

$$\mathbf{E}_g = \mathcal{P}_g(\mathbf{X}') \left(\mathbf{W}_e^{(g)}\right)^\top + \mathbf{1} \left(\mathbf{b}_e^{(g)}\right)^\top \in \mathbb{R}^{T \times D} \tag{4}$$

with independent parameters $\mathbf{W}_e^{(g)} \in \mathbb{R}^{D \times p^2 C_g}$ and $\mathbf{b}_e^{(g)} \in \mathbb{R}^D$. Stacking yields $\mathbf{E} = [\mathbf{E}_1^\top, \ldots, \mathbf{E}_G^\top]^\top \in \mathbb{R}^{N \times D}$, forming a block-diagonal mapping that preserves intergroup separation while reducing parameter coupling. To encode spatial layout and spectral-group identity, we add

$$\mathbf{Z}_{i,:} = \mathbf{E}_{i,:} + \mathbf{P}^{\text{spatial}}_{t,:} + \mathbf{P}^{\text{group}}_{g,:} \tag{5}$$

where $i \mapsto (g, t)$, $\mathbf{P}^{\text{spatial}} \in \mathbb{R}^{T \times D}$ encodes the 2-D spatial coordinates of patch t using sinusoidal functions, and $\mathbf{P}^{\text{group}} \in$

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$\mathbb{R}^{G\times D}$ encodes the 1-D spectral group index g in the same manner.

The resulting sequence $\mathbf{Z} \in \mathbb{R}^{N\times D}$ integrates both spatial and spectral priors and is passed to the shared encoder for further processing.

4) Group-Wise Masking: Masking is sampled independently across spectral groups, so that at a fixed spatial location, some groups may be masked while others remain visible. This cross-group asynchrony at identical spatial patches encourages the encoder to infer missing spectra from neighboring bands while retaining spatial context. Let T denote the number of spatial patches per group. For group g, we first generate a random vector

$$u_g = \left(u_{g,1}, \ldots, u_{g,T}\right), \quad u_{g,t} \sim \mathcal{U}(0,1) \text{ i.i.d} \tag{6}$$

where each entry is independently drawn from the uniform distribution on $(0, 1)$. Sorting u_g in ascending order yields a permutation of patch indices

$$\pi_g = \operatorname{argsort}\left(u_g\right) \in S_T \tag{7}$$

where S_T denotes the set of all permutations of $\{1, \ldots, T\}$.

Given a mask ratio $\tau \in (0, 1)$, the number of visible tokens per group is

$$T_{\text{keep}} = \lfloor (1-\tau)\, T \rfloor . \tag{8}$$

We retain the first T_{keep} indices of π_g as the visible set

$$V_g = \left\{\pi_g(1), \ldots, \pi_g\left(T_{\text{keep}}\right)\right\}. \tag{9}$$

The final binary mask matrix $M \in \{0, 1\}^{G\times T}$ is then defined by the convention

$$M_{g,t} = \begin{cases} 0, & t \in V_g \quad \text{(visible)} \\ 1, & t \notin V_g \quad \text{(masked)} \end{cases} \tag{10}$$

so that visible tokens ($M = 0$) are passed to the encoder, while masked tokens ($M = 1$) are replaced by mask tokens during reconstruction. In practice, a higher mask ratio τ_h is used for the reconstruction branch, while a lower ratio τ_l is applied for contrastive view generation, both relying on this same group-wise masking scheme.

C. Shared Encoder and Dual-Task Architecture

The proposed framework adopts a partially shared encoder and a dual-task architecture. This design ensures complementary representation learning, where one branch focuses on capturing fine-grained spectral–spatial patterns while the other enforces global discriminability among samples.

1) Shared Encoder: Given the input embedding sequence $\mathbf{Z}_{\text{visible}}$ obtained under the task-specific mask M, the shared encoder $\mathcal{E}_s$ extracts a common spectral–spatial representation $\mathbf{F}_s$, which serves as the shared feature space for both downstream objectives. The reconstruction adapter $\mathcal{E}_{\text{rec}}$ and the contrastive adapter $\mathcal{E}_{\text{con}}$ further process $\mathbf{F}_s$ to produce task-specific representations

$$\mathbf{F}_s = \mathcal{E}_s\left(\mathbf{Z}_{\text{visible}}\right), \mathbf{F}_{\text{rec}} = \mathcal{E}_{\text{rec}}\left(\mathbf{F}_s\right), \mathbf{F}_{\text{con}} = \mathcal{E}_{\text{con}}\left(\mathbf{F}_s\right). \tag{11}$$

Both branches operate on the same $\mathbf{F}_s$, ensuring that reconstruction and contrastive objectives are optimized in a common representation space.

2) Reconstruction Task: For the reconstruction task, a group-wise mask $M^{\text{rec}} \in \{0, 1\}^{G\times T}$ is applied. The decoder receives the encoded visible features $\mathbf{F}_{\text{rec}}$ together with learnable mask tokens and predicts the original patchified group-patch targets. Let $\mathcal{P}(\mathbf{X}') \in \mathbb{R}^{N\times p^2 C_g}$ be the ground-truth patch targets and $\hat{\mathcal{P}}$ the predicted ones. The reconstruction loss is computed only on masked positions

$$\mathcal{L}_{\text{rec}} = \frac{1}{\sum_{g,t} M^{\text{rec}}_{g,t}} \sum_{g,t} M^{\text{rec}}_{g,t} \left\| \hat{\mathcal{P}}_{g,t} - \mathcal{P}_{g,t}\left(\mathbf{X}'\right) \right\|_2^2. \tag{12}$$

The decoder is asymmetric and intentionally lightweight: it projects encoded visible features to a lower dimensional decoder embedding of dimension $D_{\text{dec}} < D$ and applies L_{dec} transformer blocks with reduced width and fewer attention heads than the encoder. This design reduces compute, mitigates shortcut solutions that may arise with an overly expressive decoder, and encourages the shared encoder to encode informative visible-context features.

3) Contrastive Task: For the contrastive task, two correlated views are generated by sampling two independent masks $M^{(1)}$ and $M^{(2)}$ with ratio τ_l, without applying additional augmentations. The shared encoder and contrastive branch produce features $\mathbf{F}^{(1)}_{\text{con}}$ and $\mathbf{F}^{(2)}_{\text{con}}$ for the two views. Group-level descriptors $\mathbf{v}^{(k)}$ for each view are obtained by mean-pooling spatial tokens within each group and concatenating them

$$\mathbf{v}^{(k)} = \operatorname{concat}_{g=1}^{G} \left(\frac{1}{|S_g^{(k)}|} \sum_{t\in S_g^{(k)}} \mathbf{f}^{(k)}_{c,g,t} \right) \tag{13}$$

where $S_g^{(k)}$ denotes the set of visible spatial indices for group g in view k. After projection via a multilayer perceptron (projector) and ℓ_2 normalization, the descriptors are compared across views.

With a batch of B samples, let the normalized projections form $\mathbf{Z} \in \mathbb{R}^{2B\times d}$ and define the similarity matrix $\mathbf{S} = \mathbf{Z}\mathbf{Z}^{\top}/\tau_{\text{tmp}}$, where τ_{tmp} denotes the temperature. The positive index $p(i)$ pairs two views of the same sample. The contrastive loss (NT-Xent) is defined as

$$\mathcal{L}_{\text{con}} = -\frac{1}{2B} \sum_{i=1}^{2B} \log \frac{\exp\left(\mathbf{S}_{i,p(i)}\right)}{\sum_{\substack{j=1 \\ j\neq i}}^{2B} \exp\left(\mathbf{S}_{i,j}\right)}. \tag{14}$$

The overall pretraining objective combines the two tasks via a convex combination

$$\mathcal{L}_{\text{total}} = \lambda\, \mathcal{L}_{\text{rec}} + (1-\lambda)\, \mathcal{L}_{\text{con}} \tag{15}$$

where $\lambda \in [0, 1]$ controls the tradeoff between reconstruction and contrastive objectives, and keeps the overall loss magnitude on a comparable scale when adjusting the balance coefficient. All parameters of the encoder, branches, decoder, and projector are optimized end-to-end.

D. Rationale and Downstream Adaptation

The dual-task design of MSG is intended to learn representations that are both locally precise and globally discriminative. The reconstruction objective enforces fine-grained spectral–spatial fidelity, while the contrastive objective

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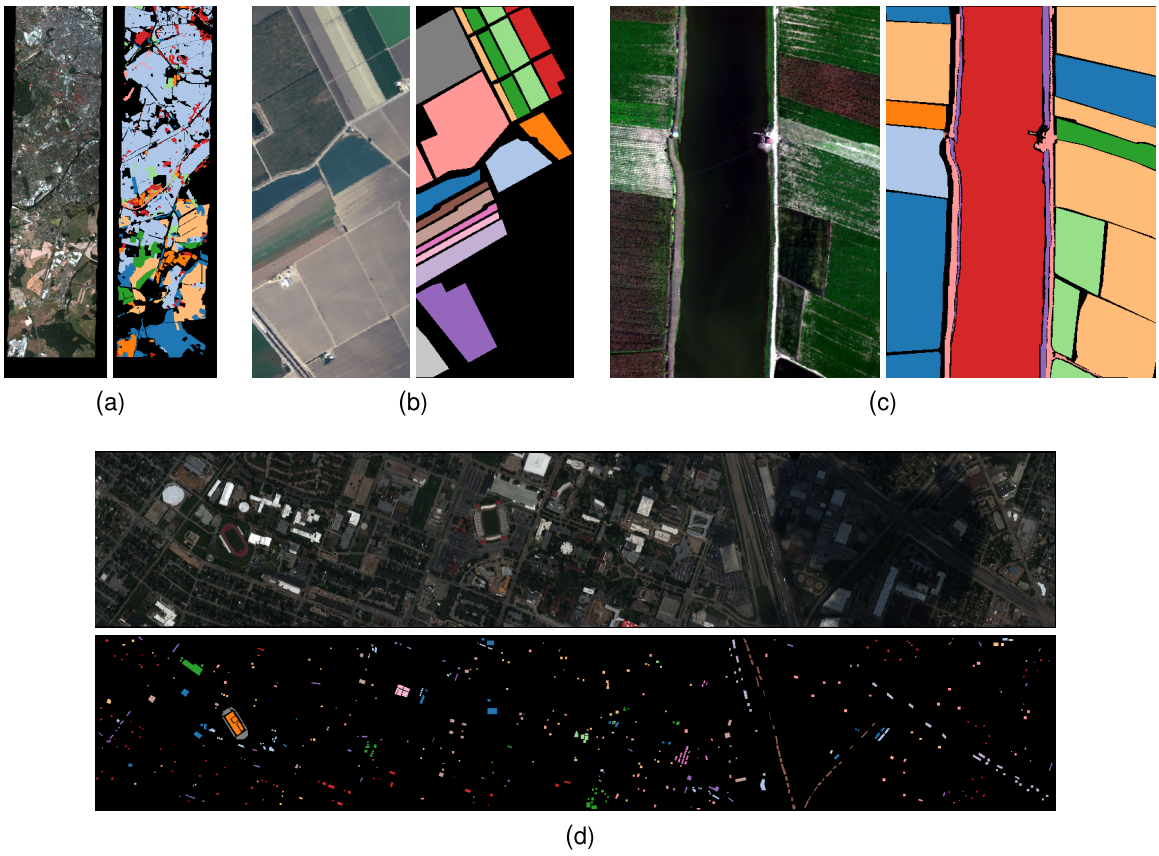


Fig. 2. Visualization of benchmark hyperspectral datasets used for downstream evaluation: (a) Berlin, (b) Salinas, (c) WHU-Hi-LongKou, and (d) Houston2013. For (a)–(c), the left shows an RGB composite generated from selected spectral bands, while the right displays the corresponding ground-truth map. For Houston2013 in (d), the RGB composite is shown on the top and the corresponding ground-truth map is shown below. All images are normalized for clarity of visualization, and the spatial coverage and aspect ratio differ across datasets.

promotes invariance to perturbations and improves the linear separability of features in the latent space by contrasting different views. The joint optimization of these complementary objectives enables the shared encoder to learn a compact, robust, and transferable representation space.

After pretraining, the decoder and the projector are removed, and the pretrained backbone consisting of SGEM and the shared encoder can be flexibly adapted to a variety of downstream settings. Since the learned features are group-aware and capture both spectral continuity and spatial structure, they provide a stable basis that can be used with different adaptation strategies depending on label availability, computational budgets, and the complexity of the target task. Lightweight adaptation can exploit these pretrained features with minimal parameter updates, while more comprehensive tuning allows the model to specialize to specific application domains.

Moreover, the structured design of MSG facilitates the attachment of diverse task heads and aggregation schemes. For example, group-level descriptors can be concatenated or pooled for image-level classification, whereas token-level outputs can be preserved before group pooling to support dense prediction tasks such as segmentation or anomaly detection. This flexibility makes MSG suitable not only as a self-supervised pretraining framework but also as a general-purpose backbone for a broad range of hyperspectral applications.

IV. Experiments and Results

A. Dataset Description

To comprehensively evaluate the proposed MSG framework, HySpecNet-11k is used for large-scale self-supervised pretraining, whereas four publicly available labeled benchmarks—Berlin, Salinas, WHU-Hi-LongKou, and Houston2013—are used for downstream evaluation. These datasets span different sensors, spatial resolutions, and application domains, thereby enabling a fair and generalizable evaluation of the learned representations. Representative visualizations of the four labeled datasets are provided in Fig. 2, and detailed category statistics are summarized in Table I.

1) HySpecNet-11k: The HySpecNet-11k dataset is a large-scale benchmark acquired by the Environmental Mapping and Analysis Program (EnMAP) satellite during routine operations in November 2022 [52]. It contains 11 483 nonoverlapping image patches of size 128×128 pixels, each with 224 spectral bands covering 420–2450 nm at a ground sampling distance of 30 m. After atmospheric correction, bands severely affected by water vapor absorption were removed, leaving 202 valid bands.

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TABLE I
CATEGORY STATISTICS OF THE BENCHMARK HYPERSPECTRAL DATASETS

No.	Color	Berlin		Salinas		WHU-Hi-LongKou		Houston2013	
		Class	Samples	Class	Samples	Class	Samples	Class	Samples
1		Forest	54954	Brocoli green weeds 1	2009	Corn	34511	Healthy Grass	1251
2		Residential area	268642	Brocoli green weeds 2	3726	Cotton	8374	Stressed Grass	1254
3		Industrial area	19566	Fallow	1976	Sesame	3031	Synthetic Grass	697
4		Low plants	59282	Fallow rough plow	1394	Broad-leaf soybean	63212	Trees	1244
5		Soil	17426	Fallow smooth	2678	Narrow-leaf soybean	4151	Soil	1242
6		Allotment	13305	Stubble	3959	Rice	11854	Water	325
7		Commercial area	24824	Celery	3579	Water	67056	Residential	1268
8		Water	6672	Grapes untrained	11271	Roads and houses	7124	Commercial	1244
9				Soil vineyard develop	6203	Mixed weed	5229	Road	1252
10				Corn senesced green weeds	3278			Highway	1227
11				Lettuce romaine 4wk	1068			Railway	1235
12				Lettuce romaine 5wk	1927			Parking Lot 1	1233
13				Lettuce romaine 6wk	916			Parking Lot 2	469
14				Lettuce romaine 7wk	1070			Tennis Court	428
15				Vineyard untrained	7268			Running Track	660
16				Vineyard vertical trellis	1807				
Total			**464671**		**54129**		**204542**		**28232**

The dataset spans diverse geographic regions and land-cover types, providing a large-scale corpus tailored for unsupervised and self-supervised representation learning.

2) Berlin: The Berlin dataset originates from the DLR EnMAP preparatory flight campaign conducted in 2009 [53]. Several flight lines were acquired using the HyMap airborne sensor over the Berlin-urban-gradient region. In this work, we use the subset with dimensions $1723 \times 476 \times 244$, which includes 244 spectral bands covering approximately 450–2500 nm with a nominal spatial resolution of 4 m. The dataset captures a distinctive urban-to-rural gradient, including built-up areas, vegetation, bare soil, and water bodies, making it a valuable benchmark for hyperspectral classification in heterogeneous urban landscapes.

3) Salinas: The Salinas dataset was collected by the AVIRIS sensor over the agricultural area of Salinas Valley, California. It consists of 512×217 pixels with 224 spectral bands covering 400–2500 nm at a spatial resolution of 3.7 m. After removing 20 water absorption bands, 204 bands remain. The dataset contains 16 land-cover classes, primarily corresponding to different crop types such as broccoli, lettuce, and vineyards. Owing to its high intraclass variability and large number of categories, Salinas serves as a challenging benchmark widely adopted in HSI classification studies.

4) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset is part of the WHU-Hi UAV-borne benchmark suite, which was designed for precise crop classification [54]. It was acquired on July 17, 2018, in Longkou Town, Hubei Province, China, using a Headwall Nano-Hyperspec sensor mounted on a DJI M600 Pro UAV platform. The data cube has dimensions of $550 \times 400 \times 270$, with spectral coverage from 400 to 1000 nm and a ground sampling distance of approximately 0.46 m. The scene contains six major crop species, including corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. With its fine spectral granularity and ultrahigh spatial resolution, WHU-Hi-LongKou provides a representative benchmark for evaluating hyperspectral classification methods in smallholder agricultural environments.

5) Houston2013: The Houston2013 dataset was released as part of the IEEE GRSS Data Fusion Contest 2013 and has become a widely used benchmark for urban HSI classification [55]. It was acquired by the ITRES CASI-1500 airborne hyperspectral sensor over the University of Houston campus and its surrounding urban areas. The data cube has a spatial size of 349×1905 pixels with a spatial resolution of 2.5 m, covering the spectral range from 380 to 1050 nm with 144 spectral bands. The scene represents a complex urban environment with high spectral variability and strong spatial heterogeneity, including materials such as roads, buildings, parking lots, grass, trees, water, and residential areas. A total of 15 land-cover classes are annotated, making the dataset particularly suitable for evaluating the robustness and generalization ability of hyperspectral classification and representation learning methods under challenging urban scenarios.

B. Experimental Setup

1) Evaluation Metrics: We adopt three widely used metrics for HSI classification: overall accuracy (OA), average accuracy (AA), and Cohen's kappa coefficient (κ). These measures provide complementary perspectives on model performance across different classes.

Given a confusion matrix $\mathbf{M} \in \mathbb{N}^{n_c \times n_c}$ with n_c classes and a total of N samples, OA is defined as

$$\mathrm{OA} = \frac{\sum_{i=1}^{n_c} M_{ii}}{N}. \tag{16}$$

The per-class accuracy for class i is defined as

$$a_i = \frac{M_{ii}}{\sum_{j=1}^{n_c} M_{ij}} \tag{17}$$

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TABLE II
LINEAR EVALUATION RESULTS ON BERLIN DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	83.53 ± 0.83	77.76 ± 1.60	82.82 ± 0.86	82.82 ± 2.92	80.94 ± 1.09	83.76 ± 1.88	**88.00 ± 1.52**
2	91.86 ± 0.41	89.95 ± 0.41	90.27 ± 0.81	90.65 ± 1.88	**95.19 ± 0.28**	93.21 ± 0.78	90.94 ± 0.46
3	59.38 ± 2.21	29.69 ± 5.23	62.81 ± 2.30	65.00 ± 3.78	32.19 ± 2.12	37.81 ± 8.75	**67.81 ± 3.06**
4	87.33 ± 0.65	85.89 ± 0.75	86.33 ± 0.27	89.67 ± 1.20	85.11 ± 0.82	86.11 ± 0.99	**90.00 ± 1.76**
5	84.53 ± 0.75	79.25 ± 2.07	86.79 ± 1.19	89.06 ± 2.77	75.85 ± 0.75	90.19 ± 1.41	**90.57 ± 4.77**
6	41.46 ± 1.54	12.68 ± 2.84	44.88 ± 4.25	49.76 ± 8.39	4.39 ± 3.58	37.07 ± 9.80	**72.20 ± 5.48**
7	27.20 ± 3.33	14.93 ± 4.73	28.27 ± 4.17	33.60 ± 8.90	18.40 ± 2.59	24.53 ± 4.18	**45.33 ± 6.90**
8	54.55 ± 0.00	66.36 ± 12.40	80.91 ± 4.45	83.64 ± 8.43	46.36 ± 8.33	71.82 ± 13.30	**94.55 ± 1.82**
OA (%)	83.18 ± 0.17	78.41 ± 0.62	82.85 ± 0.32	84.13 ± 0.49	81.34 ± 0.21	83.08 ± 0.67	**86.56 ± 0.21**
AA (%)	66.23 ± 0.54	57.06 ± 2.18	70.39 ± 0.89	73.02 ± 2.69	54.80 ± 1.41	65.56 ± 3.47	**79.92 ± 1.00**
Kappa (×100)	72.37 ± 0.29	64.13 ± 1.17	72.38 ± 0.41	74.50 ± 0.72	67.89 ± 0.43	71.76 ± 1.37	**78.80 ± 0.43**

TABLE III
LINEAR EVALUATION RESULTS ON SALINAS DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
2	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
3	6.67 ± 8.16	**100.00 ± 0.00**	20.00 ± 6.67	70.00 ± 12.47	56.67 ± 8.16	93.33 ± 8.16	**100.00 ± 0.00**
4	92.00 ± 16.00	76.00 ± 8.00	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
5	**100.00 ± 0.00**	**100.00 ± 0.00**	95.00 ± 6.12	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
6	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
7	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	**100.00 ± 0.00**	86.86 ± 2.29	92.57 ± 5.30	83.43 ± 3.33	88.57 ± 1.81	78.86 ± 8.00	84.00 ± 5.60
9	**100.00 ± 0.00**	96.84 ± 2.58	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
10	60.00 ± 4.45	54.55 ± 0.00	58.18 ± 7.27	65.45 ± 6.80	72.73 ± 0.00	76.36 ± 4.45	**92.73 ± 3.64**
11	0.00 ± 0.00	46.67 ± 26.67	13.33 ± 16.33	86.67 ± 16.33	**100.00 ± 0.00**	93.33 ± 13.33	**100.00 ± 0.00**
12	0.00 ± 0.00	43.33 ± 17.00	70.00 ± 6.67	83.33 ± 0.00	93.33 ± 13.33	96.67 ± 6.67	**100.00 ± 0.00**
13	0.00 ± 0.00	**100.00 ± 0.00**	6.67 ± 13.33	66.67 ± 0.00	93.33 ± 13.33	93.33 ± 13.33	**100.00 ± 0.00**
14	66.67 ± 0.00	**100.00 ± 0.00**	86.67 ± 16.33	93.33 ± 13.33	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
15	5.83 ± 4.25	37.50 ± 4.56	15.83 ± 7.17	56.67 ± 6.77	45.83 ± 10.87	74.17 ± 7.17	**84.17 ± 3.12**
16	56.67 ± 8.16	**83.33 ± 0.00**	80.00 ± 6.67	**83.33 ± 0.00**	**83.33 ± 0.00**	**83.33 ± 0.00**	**83.33 ± 0.00**
OA (%)	71.41 ± 0.80	80.94 ± 1.03	75.65 ± 0.80	85.06 ± 1.77	85.76 ± 1.36	89.29 ± 0.94	**93.41 ± 1.14**
AA (%)	61.74 ± 0.99	82.82 ± 1.78	71.14 ± 1.25	86.81 ± 2.67	89.61 ± 1.93	93.09 ± 1.06	**96.51 ± 0.34**
Kappa (×100)	67.38 ± 0.93	78.59 ± 1.16	72.49 ± 0.90	83.27 ± 2.00	84.05 ± 1.54	88.07 ± 1.03	**92.67 ± 1.26**

and AA is obtained as the macro-average across all classes

$$\mathrm{AA} = \frac{1}{n_c}\sum_{i=1}^{n_c} a_i. \tag{18}$$

The κ coefficient is used to measure the level of agreement beyond chance, where the expected agreement is given by

$$p_e = \sum_{i=1}^{n_c} \frac{\sum_{j=1}^{n_c} M_{ij}}{N} \cdot \frac{\sum_{j=1}^{n_c} M_{ji}}{N}. \tag{19}$$

It is then defined as

$$\kappa = \frac{\mathrm{OA} - p_e}{1 - p_e}. \tag{20}$$

All reported results are presented as the mean ± standard deviation over multiple random seeds to ensure statistical reliability.

2) Baseline Methods: We compare MSG with several representative self-supervised approaches under a unified linear evaluation protocol and few-shot FT. To better contextualize the results, we categorize the baselines into two groups: 1) *CV transfer* methods, which were originally developed for natural-image representation learning and are directly transferred to hyperspectral data with minimal architectural or objective changes; and 2) *HSI self-supervised* methods, which are specifically designed for hyperspectral or multisource remote sensing data and explicitly encode spectral–spatial inductive biases through tailored masking strategies, factorized spectral–spatial modeling, or dual-branch designs. houston These two groups correspond to the column headers in Tables II–IX.

a) MAE [26]: The original MAE adapted to hyperspectral data with adjusted hyperparameters. It partitions the input into spatial patches, applies a high masking ratio, and

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TABLE IV
LINEAR EVALUATION RESULTS ON WHU-HI-LONGKOU DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	98.87 ± 0.38	99.06 ± 0.00	99.25 ± 0.71	**100.00 ± 0.00**	99.06 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**
2	90.37 ± 1.81	75.56 ± 3.78	97.04 ± 2.77	99.26 ± 1.48	92.59 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**
3	88.89 ± 0.00	82.22 ± 5.44	84.44 ± 5.44	**100.00 ± 0.00**	88.89 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**
4	92.28 ± 0.38	93.10 ± 0.41	94.21 ± 0.69	**98.68 ± 0.94**	97.87 ± 0.81	98.48 ± 0.32	**98.68 ± 0.94**
5	85.00 ± 3.33	83.33 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
6	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
7	99.51 ± 0.00	**100.00 ± 0.00**	99.51 ± 0.00	99.80 ± 0.24	**100.00 ± 0.00**	**100.00 ± 0.00**	99.71 ± 0.24
8	92.31 ± 0.00	92.31 ± 0.00	88.46 ± 4.21	95.38 ± 1.54	**96.15 ± 0.00**	90.00 ± 3.92	94.62 ± 1.88
9	86.67 ± 0.00	86.67 ± 4.22	82.67 ± 5.33	94.67 ± 4.99	80.00 ± 0.00	90.67 ± 3.27	**100.00 ± 0.00**
OA (%)	95.77 ± 0.18	95.45 ± 0.39	96.68 ± 0.26	99.18 ± 0.31	98.07 ± 0.25	98.89 ± 0.20	**99.27 ± 0.38**
AA (%)	92.66 ± 0.38	90.25 ± 1.16	93.95 ± 1.26	98.64 ± 0.73	94.95 ± 0.09	97.68 ± 0.54	**99.22 ± 0.30**
Kappa (×100)	94.49 ± 0.24	94.05 ± 0.51	95.67 ± 0.35	98.92 ± 0.40	97.47 ± 0.33	98.55 ± 0.26	**99.05 ± 0.50**

TABLE V
LINEAR EVALUATION RESULTS ON HOUSTON2013 DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	93.75 ± 0.00	93.75 ± 0.00	92.50 ± 1.67	91.25 ± 0.83	93.75 ± 0.00	90.83 ± 3.86	**99.58 ± 0.83**
2	77.08 ± 1.32	82.92 ± 0.83	70.83 ± 5.43	83.75 ± 0.83	77.92 ± 1.67	82.50 ± 1.67	**97.50 ± 1.56**
3	24.62 ± 5.22	88.46 ± 0.00	42.31 ± 18.37	70.77 ± 11.04	95.38 ± 1.54	92.31 ± 0.00	**100.00 ± 0.00**
4	82.08 ± 2.12	89.58 ± 2.28	77.50 ± 3.58	93.33 ± 1.56	93.75 ± 0.00	86.25 ± 3.86	**96.67 ± 1.02**
5	94.89 ± 1.04	**100.00 ± 0.00**	92.77 ± 1.04	97.45 ± 4.13	98.30 ± 0.85	94.04 ± 5.77	99.15 ± 1.04
6	29.09 ± 3.64	69.09 ± 4.45	16.36 ± 6.80	69.09 ± 4.45	72.73 ± 0.00	58.18 ± 10.91	**96.36 ± 4.45**
7	52.34 ± 5.14	69.36 ± 3.95	77.02 ± 2.48	76.17 ± 6.08	76.17 ± 0.85	67.23 ± 5.14	**80.85 ± 3.56**
8	58.30 ± 1.04	49.79 ± 3.71	31.49 ± 2.48	54.04 ± 1.70	47.23 ± 2.08	46.81 ± 6.99	**62.98 ± 1.04**
9	62.80 ± 6.01	75.60 ± 3.44	85.20 ± 2.40	64.80 ± 4.66	76.00 ± 1.26	52.40 ± 5.43	**88.00 ± 5.51**
10	20.87 ± 9.18	25.22 ± 12.79	0.00 ± 0.00	44.78 ± 20.70	32.61 ± 3.64	9.13 ± 1.63	**60.87 ± 6.87**
11	56.67 ± 8.48	50.42 ± 5.80	64.17 ± 5.50	47.50 ± 18.04	60.42 ± 4.37	62.50 ± 3.23	**77.92 ± 2.12**
12	14.22 ± 4.12	38.22 ± 5.69	17.78 ± 2.43	47.11 ± 5.33	38.22 ± 2.59	31.11 ± 3.14	**76.89 ± 5.18**
13	0.00 ± 0.00	0.00 ± 0.00	72.94 ± 4.71	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**84.71 ± 7.98**
14	58.67 ± 6.53	93.33 ± 0.00	0.00 ± 0.00	89.33 ± 3.27	93.33 ± 0.00	85.33 ± 4.99	**100.00 ± 0.00**
15	**100.00 ± 0.00**	77.60 ± 4.80	96.00 ± 0.00	91.20 ± 2.99	**100.00 ± 0.00**	90.40 ± 5.43	**100.00 ± 0.00**
OA (%)	59.15 ± 0.63	67.89 ± 1.19	60.00 ± 0.49	69.54 ± 0.95	70.88 ± 0.45	63.84 ± 0.88	**86.30 ± 0.77**
AA (%)	55.03 ± 0.72	66.89 ± 1.26	55.79 ± 0.72	68.04 ± 0.94	70.39 ± 0.33	63.27 ± 1.49	**88.10 ± 0.97**
Kappa (×100)	55.68 ± 0.68	65.16 ± 1.29	56.56 ± 0.56	67.00 ± 1.03	68.47 ± 0.49	60.87 ± 0.96	**85.18 ± 0.84**

TABLE VI
FT RESULTS ON BERLIN DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	72.84 ± 5.17	46.48 ± 38.10	73.95 ± 1.06	79.71 ± 1.15	69.07 ± 3.36	77.35 ± 3.91	**81.88 ± 2.98**
2	92.34 ± 4.44	**95.83 ± 3.47**	93.78 ± 1.22	94.83 ± 0.85	93.46 ± 1.30	90.26 ± 2.89	92.45 ± 1.42
3	9.26 ± 9.62	12.05 ± 14.14	12.00 ± 6.40	25.37 ± 2.40	18.47 ± 5.49	29.42 ± 11.96	**45.42 ± 3.88**
4	81.57 ± 6.37	41.94 ± 34.61	80.65 ± 2.85	82.87 ± 1.09	76.95 ± 3.36	79.11 ± 1.50	**87.47 ± 1.66**
5	77.03 ± 7.27	40.63 ± 29.69	75.39 ± 10.37	83.15 ± 0.76	69.46 ± 8.03	82.97 ± 8.10	**90.35 ± 3.93**
6	6.32 ± 4.75	2.27 ± 3.61	5.59 ± 3.66	25.43 ± 5.76	5.91 ± 4.15	25.99 ± 19.95	**39.92 ± 4.53**
7	19.64 ± 6.48	2.53 ± 4.53	18.80 ± 2.18	29.82 ± 3.84	12.04 ± 3.66	12.71 ± 10.77	**33.69 ± 1.21**
8	34.44 ± 29.99	13.08 ± 13.84	60.00 ± 4.97	64.51 ± 4.84	22.41 ± 11.56	63.01 ± 5.59	**83.16 ± 4.72**
OA (%)	77.25 ± 0.86	68.60 ± 8.75	78.50 ± 0.74	82.16 ± 0.20	76.41 ± 0.34	78.03 ± 1.44	**83.68 ± 0.53**
AA (%)	49.18 ± 6.01	31.85 ± 15.49	52.52 ± 1.15	60.71 ± 1.55	45.97 ± 1.51	57.60 ± 4.51	**69.29 ± 1.49**
Kappa (×100)	61.12 ± 0.88	34.73 ± 27.88	63.25 ± 0.94	69.93 ± 0.45	58.61 ± 0.90	63.72 ± 2.58	**73.48 ± 0.72**

reconstructs the missing pixels using a lightweight decoder, without explicitly modeling spectral dependencies. This baseline represents a reconstruction-only paradigm directly transferred from CV.

b) SimCLR [19]: A contrastive learning framework originally proposed for natural images. In our implementation, we adopt a ViT backbone consistent with other baselines and MSG, and generate augmented views using spectral- and

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TABLE VII
FT RESULTS ON SALINAS DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	74.36 ± 38.75	94.36 ± 10.05	88.72 ± 17.51	98.46 ± 3.08	98.46 ± 3.08	**99.49 ± 1.03**	**99.49 ± 1.03**
2	97.61 ± 2.60	98.81 ± 1.74	91.34 ± 7.86	**100.00 ± 0.00**	97.61 ± 1.79	99.40 ± 1.19	99.70 ± 0.60
3	25.95 ± 17.90	78.92 ± 12.01	42.16 ± 38.09	67.57 ± 10.11	51.35 ± 23.37	56.22 ± 25.20	**94.05 ± 11.89**
4	69.29 ± 36.18	87.86 ± 10.25	50.00 ± 32.65	**100.00 ± 0.00**	91.43 ± 5.80	99.29 ± 1.43	95.71 ± 8.57
5	75.42 ± 37.81	79.17 ± 17.43	64.17 ± 33.19	88.75 ± 1.67	75.00 ± 7.57	87.92 ± 3.82	**95.42 ± 1.56**
6	85.56 ± 28.89	**100.00 ± 0.00**	98.61 ± 2.78	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.17 ± 1.67
7	77.19 ± 38.67	86.56 ± 16.38	84.06 ± 15.92	99.06 ± 0.77	95.00 ± 4.00	97.19 ± 4.24	**99.38 ± 1.25**
8	**85.55 ± 8.11**	80.86 ± 13.42	74.45 ± 17.30	73.30 ± 5.86	84.88 ± 9.12	82.01 ± 5.01	80.67 ± 1.27
9	97.89 ± 2.91	98.25 ± 1.57	95.09 ± 8.53	**100.00 ± 0.00**	99.30 ± 1.40	99.65 ± 0.43	**100.00 ± 0.00**
10	70.16 ± 5.80	70.48 ± 6.24	36.83 ± 30.69	**81.90 ± 4.67**	72.38 ± 17.52	63.81 ± 17.22	79.37 ± 4.14
11	0.95 ± 1.90	16.19 ± 11.90	7.62 ± 8.83	65.71 ± 20.07	27.62 ± 23.41	5.71 ± 7.62	**90.48 ± 7.38**
12	60.54 ± 32.55	52.43 ± 21.66	35.14 ± 36.94	90.81 ± 4.39	42.16 ± 20.84	69.73 ± 11.52	**94.59 ± 4.52**
13	76.67 ± 38.39	**94.44 ± 4.97**	44.44 ± 33.88	83.33 ± 0.00	71.11 ± 12.86	91.11 ± 4.44	85.56 ± 28.89
14	72.38 ± 36.93	86.67 ± 7.62	54.29 ± 44.63	93.33 ± 2.33	82.86 ± 21.21	97.14 ± 5.71	**100.00 ± 0.00**
15	21.43 ± 18.26	41.00 ± 21.31	31.86 ± 26.41	**72.71 ± 8.82**	30.86 ± 15.99	48.14 ± 19.44	54.86 ± 6.25
16	66.86 ± 33.59	84.57 ± 6.66	66.86 ± 38.74	85.14 ± 2.80	85.14 ± 6.86	66.86 ± 9.14	**94.29 ± 0.00**
OA (%)	70.11 ± 13.50	78.24 ± 4.45	66.16 ± 10.63	85.77 ± 0.54	76.80 ± 2.45	79.98 ± 4.25	**86.95 ± 1.86**
AA (%)	66.11 ± 21.30	78.16 ± 4.79	60.35 ± 19.11	87.51 ± 0.65	75.32 ± 3.96	78.98 ± 4.65	**91.42 ± 3.56**
Kappa (×100)	65.53 ± 16.94	75.64 ± 5.05	61.61 ± 12.66	84.17 ± 0.59	73.99 ± 2.77	77.52 ± 4.87	**85.47 ± 2.06**

TABLE VIII
FT RESULTS ON WHU-HI-LONGKOU DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	98.90 ± 0.48	95.47 ± 1.78	98.55 ± 1.08	99.43 ± 0.41	97.20 ± 1.21	97.67 ± 1.25	**99.75 ± 0.21**
2	75.82 ± 7.54	61.77 ± 5.50	62.15 ± 21.37	68.61 ± 6.26	64.56 ± 9.46	73.54 ± 11.28	**90.89 ± 1.48**
3	87.14 ± 6.53	64.29 ± 11.35	56.07 ± 23.65	88.57 ± 8.57	74.64 ± 20.44	93.21 ± 3.64	**97.14 ± 2.42**
4	96.62 ± 0.82	95.15 ± 1.72	96.62 ± 0.45	97.43 ± 1.63	96.10 ± 1.35	95.62 ± 1.47	**97.89 ± 0.84**
5	67.89 ± 10.11	40.26 ± 7.33	61.32 ± 9.36	80.26 ± 7.30	57.11 ± 13.69	36.84 ± 19.21	**98.42 ± 0.53**
6	99.55 ± 0.57	96.36 ± 2.71	81.00 ± 7.85	99.09 ± 0.64	97.27 ± 3.70	92.36 ± 6.17	**99.73 ± 0.36**
7	99.82 ± 0.10	**99.98 ± 0.03**	99.84 ± 0.15	99.95 ± 0.07	99.71 ± 0.08	99.84 ± 0.05	99.61 ± 0.13
8	82.09 ± 6.06	83.27 ± 8.21	76.73 ± 14.51	92.16 ± 4.20	80.26 ± 2.96	90.33 ± 2.42	**98.95 ± 0.52**
9	43.86 ± 4.85	49.55 ± 6.08	52.50 ± 6.24	43.86 ± 10.87	45.91 ± 2.75	48.18 ± 6.04	**61.82 ± 1.16**
OA (%)	94.82 ± 0.26	92.36 ± 1.15	92.51 ± 0.36	95.55 ± 0.43	93.30 ± 0.69	93.69 ± 1.12	**97.78 ± 0.26**
AA (%)	83.52 ± 1.70	76.23 ± 3.07	76.09 ± 2.38	85.49 ± 1.12	79.19 ± 3.16	80.85 ± 3.58	**93.80 ± 0.44**
Kappa (×100)	93.18 ± 0.35	89.91 ± 1.51	90.11 ± 0.49	94.13 ± 0.56	91.18 ± 0.91	91.72 ± 1.48	**97.09 ± 0.34**

spatial-preserving transformations. This baseline represents a contrastive-only paradigm for hyperspectral data.

c) SS-MAE [56]: A spatial–spectral MAE designed for multisource remote sensing classification. It adopts a dual-branch pretraining strategy, where one branch masks spatial patches and the other masks entire spectral bands, enabling joint modeling of spatial and spectral contexts. The original design includes lightweight convolutional modules for spatial enhancement during FT, which are omitted in our experiments to ensure a fair linear evaluation across baselines.

d) HSIMAE [57]: An MAE tailored for hyperspectral data with a staged spatial–spectral masking strategy. Given a 3-D cube, spatial masking first removes all tokens at the same spatial location across spectral bands, and spectral masking subsequently removes tokens at the same spectral location across the image. This two-stage design preserves spatial and spectral consistency during masking and guides reconstruction to capture coupled spectral–spatial dependencies using a lightweight MAE-style decoder.

e) TMAC [24]: A transformer-based MAE augmented with a momentum contrastive branch for hyperspectral representation learning. The model jointly optimizes masked reconstruction and contrastive objectives, where queries are produced by a momentum encoder on visible tokens, and keys are obtained by passing the corresponding features through a lightweight decoder and a projection head. This design offers a straightforward coupling between reconstruction and contrastive learning, yet the interaction between the two objectives remains relatively shallow.

f) FactoFormer [58]: A factorized transformer pretraining framework for hyperspectral representation learning that

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TABLE IX
FT RESULTS ON HOUSTON2013 DATASET

Class	CV transfer		HSI self-supervised				Proposed
	MAE	SimCLR	SS-MAE	HSI-MAE	TMAC	FactoFormer	MSG
1	82.59 ± 1.02	81.82 ± 2.55	**98.18 ± 0.92**	98.11 ± 0.57	96.78 ± 2.48	94.13 ± 3.70	96.22 ± 0.34
2	76.89 ± 5.31	83.11 ± 5.16	85.51 ± 5.38	79.36 ± 3.93	74.42 ± 6.17	67.70 ± 2.27	**87.21 ± 0.14**
3	68.68 ± 8.42	89.31 ± 5.35	82.77 ± 6.78	95.22 ± 5.84	93.71 ± 2.29	89.94 ± 9.49	**99.75 ± 0.31**
4	91.83 ± 2.39	87.11 ± 4.49	91.76 ± 2.07	91.76 ± 1.19	91.55 ± 1.39	91.83 ± 2.00	**94.86 ± 0.57**
5	81.28 ± 1.61	90.28 ± 3.90	95.67 ± 2.71	99.57 ± 0.41	97.87 ± 0.39	82.06 ± 1.45	**100.00 ± 0.00**
6	0.00 ± 0.00	62.94 ± 11.96	72.35 ± 5.53	80.88 ± 1.61	77.06 ± 5.39	60.88 ± 7.30	**85.29 ± 0.00**
7	78.79 ± 3.17	72.07 ± 8.86	58.93 ± 9.37	77.79 ± 3.75	79.93 ± 3.13	59.71 ± 11.46	**89.71 ± 0.80**
8	49.75 ± 2.76	51.60 ± 4.71	**66.69 ± 4.39**	63.27 ± 1.61	63.49 ± 5.93	46.62 ± 4.30	61.14 ± 1.53
9	73.54 ± 6.17	63.43 ± 6.22	73.47 ± 7.46	**80.20 ± 1.43**	79.73 ± 1.87	61.14 ± 6.03	78.79 ± 1.72
10	16.19 ± 7.34	8.13 ± 6.09	64.10 ± 2.10	79.14 ± 3.44	77.48 ± 6.65	31.87 ± 13.76	**88.56 ± 1.74**
11	59.30 ± 5.06	51.20 ± 10.75	70.07 ± 6.78	65.56 ± 4.74	67.89 ± 2.29	53.10 ± 5.10	**84.44 ± 0.68**
12	37.13 ± 5.94	18.97 ± 8.62	64.93 ± 4.04	69.63 ± 5.31	73.53 ± 5.17	48.60 ± 15.43	**77.65 ± 1.05**
13	46.79 ± 34.38	0.00 ± 0.00	45.47 ± 5.18	**92.83 ± 2.03**	85.66 ± 6.65	37.17 ± 9.41	80.38 ± 0.71
14	87.58 ± 6.60	96.84 ± 4.75	90.74 ± 4.34	99.16 ± 0.79	92.21 ± 3.02	93.26 ± 1.07	**100.00 ± 0.00**
15	90.13 ± 11.85	11.20 ± 14.63	89.07 ± 10.48	99.07 ± 0.53	97.87 ± 0.50	94.80 ± 2.04	**100.00 ± 0.00**
OA (%)	65.03 ± 2.94	59.31 ± 1.59	77.13 ± 0.97	82.93 ± 0.99	82.15 ± 1.16	66.34 ± 1.35	**87.35 ± 0.34**
AA (%)	62.70 ± 3.96	57.87 ± 1.86	76.65 ± 0.88	84.77 ± 0.89	83.28 ± 1.41	67.52 ± 1.41	**88.27 ± 0.26**
Kappa (×100)	62.18 ± 3.21	56.01 ± 1.72	75.26 ± 1.04	81.54 ± 1.07	80.70 ± 1.26	63.75 ± 1.46	**86.33 ± 0.37**

explicitly decouples spatial and spectral tokenization. It performs two-stage pretraining: a spatial branch partitions the input cube into spatial patches and reconstructs masked patch content, while a spectral branch treats each band as a token and reconstructs masked band-wise responses. The two pretrained encoders are then jointly transferred to downstream tasks by concatenating their CLS representations and training a lightweight classification head.

3) Implementation Details: All methods follow a consistent preprocessing strategy to ensure fairness. GW-PCA is first applied to reduce spectral redundancy and project each hyperspectral patch into 128 channels. Afterward, the processed data is divided into 7×7 spatial patches, each of which is tokenized with a patch size set to 1. Based on this representation, the SGEM module partitions the 128 channels into eight groups to generate group-wise tokens. The encoder consists of six layers with an embedding dimension of 128 and eight attention heads, while the decoder is lightweight with four layers and a hidden dimension of 64. Unless otherwise stated, the masking ratio is set to 0.7 in the reconstruction branch and 0.3 in the contrastive branch. Contrastive representations are mainly obtained by applying low-ratio masking, supplemented with mild Gaussian noise to introduce variability while preserving spectral continuity. The contrastive loss uses a balancing weight of 1.0.

Pretraining runs for 200 epochs using the AdamW optimizer with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-2}. A cosine learning-rate schedule with five warmup epochs is adopted. The batch size is fixed at 128, and up to 100 000 hyperspectral patches are sampled for each run. Mixed-precision training with gradient scaling is used to improve efficiency. An early stopping criterion with a patience of 20 epochs is applied, and the model with the lowest validation loss is retained.

For downstream experiments, two evaluation protocols are considered: linear evaluation and FT. For both protocols, the crop size is fixed at 7×7, and a stride of 7 is used by default to avoid overlap between adjacent samples. The only exception is the Houston dataset in the FT setting, where the stride is reduced to 2 because a stride of 7 would yield insufficient training samples under the 5% label ratio. In linear evaluation, all pretrained weights are frozen, and only a single linear classifier is trained on the concatenated features from the reconstruction and contrastive branches using the Adam optimizer with a learning rate of 1×10^{-3}. In FT, all backbone parameters are unfrozen and jointly optimized with the classifier under the same learning rate, while the weight decay is set to 0 to encourage rapid adaptation in the low-resource regime. Notably, all downstream experiments are conducted on datasets that are strictly disjoint from the pretraining data, ensuring a rigorous and unbiased assessment of cross-dataset transferability.

All experiments are implemented in PyTorch and executed on a system equipped with an NVIDIA GeForce RTX 5070Ti graphics card with 16-GB memory and 32-GB RAM. All reported results are averaged over five different random seeds and presented as mean ± standard deviation to ensure statistical reliability.

C. Performance Comparison

1) Linear Evaluation: We assess representation quality under a standard linear evaluation protocol. All backbone parameters are frozen, and only a linear classifier is trained on top of the pretrained features. For architectures with two branches, their frozen outputs are concatenated before the classifier; single-branch methods use the encoder output directly. We adopt a 70/15/15 split for train/validation/test and enforce nonoverlapping patch sampling to avoid information leakage

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for Berlin, Salinas, and WHU-Hi-LongKou. For Houston2013, due to its limited and spatially sparse labeled samples, a small amount of overlap may occur during dispersed patch sampling. This setting isolates the intrinsic discriminative power of the learned features by removing the confounding effects of downstream FT.

As summarized in Tables II–V, MSG achieves the best overall performance in terms of OA, AA, and κ across all four benchmarks, outperforming both CV transfer and HSI self-supervised methods. On Berlin, MSG reaches OA = 86.56%, AA = 79.92%, and κ = 78.80%. It performs particularly well on structurally complex categories such as industrial area (Class 3), commercial area (Class 7), and water (Class 8), where accurate classification is challenging due to strong spectral–spatial mixing. On Salinas, MSG obtains OA = 93.41%, AA = 96.51%, and κ = 92.67%, achieving nearly perfect accuracy on lettuce romaine categories (Classes 11–14) and stable performance on corn senesced green weeds (Class 10), which involves subtle spectral differences. On WHU-Hi-LongKou, the task is near-saturated, yet MSG still secures the best aggregate results with OA = 99.27%, AA = 99.22%, and κ = 99.05%, reaching 100% accuracy on corn (Class 1) and maintaining strong stability on roads and houses (Class 8) and mixed weed (Class 9). On Houston2013, MSG achieves OA = 86.30%, AA = 88.10%, and κ = 85.18%, showing consistent gains on urban categories with high intraclass variability and confusing materials, indicating more robust discriminative representations.

These results show that the proposed model learns highly discriminative and separable features. MSG exhibits not only strong performance on dominant categories but also robustness on structurally or spectrally challenging classes, reflecting its ability to capture both global and local structure effectively during pretraining.

2) Fine-Tuning: To further examine the transferability and robustness of the learned features, we conduct end-to-end FT. A 5/5/90 split is adopted to emphasize small-sample adaptation, with the same sampling strategy. Unlike linear evaluation, the backbone parameters are unfrozen to allow full network optimization. This setting evaluates whether MSG provides a favorable initialization that accelerates and stabilizes downstream adaptation.

As shown in Tables VI–IX, MSG consistently achieves the best OA, AA, and κ on all datasets, further confirming its effectiveness. On the Berlin dataset, it achieves OA = 83.68%, AA = 69.29%, and κ = 73.48%, with stable accuracy on industrial area (Class 3) and water (Class 8), which are prone to error under small-sample conditions. On Salinas dataset, MSG obtains OA = 86.95%, AA = 91.42%, and κ = 85.47%, showing clear advantages on fallow (Class 3) and lettuce romaine 4wk (Class 11), which typically exhibit performance degradation in low-data settings. On the WHU-Hi-LongKou dataset, MSG achieves OA = 97.78%, AA = 93.80%, and κ = 97.09%, maintaining strong accuracy on both major classes like corn (Class 1) and water (Class 7), and more difficult categories such as mixed weed (Class 9). On the Houston2013 dataset, MSG achieves OA = 87.35%, AA = 88.27%, and κ = 86.33%, with clear gains on several confusing categories such as water (Class 6, 85.29%), highway (Class 10, 88.56%), and parking lot 1 (Class 12, 77.65%). It also reaches 100.00% on soil (Class 5), tennis court (Class 14), and running track (Class 15), indicating stable small-sample adaptation in complex urban scenes.

This experiment highlights the adaptability of MSG under small-sample supervision. More importantly, since the pretraining and FT datasets are disjoint, the results also demonstrate the strong cross-dataset transferability of MSG, showing that the learned representations generalize well to new sensors, regions, and land-cover distributions.

D. Ablation Study and Analysis

1) Component Ablation: To better understand the contribution of each component to the overall performance of MSG, we conduct a series of ablation experiments under the linear evaluation protocol. These experiments are designed to disentangle the effects of masking strategy, learning objective, and architectural design, thereby offering a more granular view of how different elements interact to shape representation quality.

Seven experimental groups are configured by incrementally enabling or disabling specific components. Groups 1 and 2 follow the original MAE paradigm with spatial and band-wise masking, respectively, to examine the effectiveness of different masking strategies. Groups 3 and 4 adopt spectral-group masking while retaining only the reconstruction or contrastive branch, respectively, to isolate and analyze the contribution of each learning objective. Groups 5 and 6 keep both branches but modify the architecture. Group 5 removes the spectral stem to assess its role as an initial projection module, whereas Group 6 employs separate encoders (sep. enc.), meaning that the two branches are optimized independently without interaction. Finally, Group 7 corresponds to the full MSG model, which integrates spectral-group masking, dual learning objectives, the spectral stem, and a shared encoder. For clarity, we report OA in percentage for all datasets.

The results summarized in Table X clearly demonstrate the contribution of each component. Groups 1 and 2, which rely on conventional spatial or band-wise masking, yield the lowest OA, highlighting the limitations of these simple strategies for capturing spectral–spatial dependencies. By contrast, Groups 3 and 4 achieve substantial OA improvements, confirming that spectral-group masking provides stronger structural priors even when used with a single learning objective. Moreover, the OA gap between these single-objective variants and Group 7 indicates that reconstruction and contrastive learning are complementary rather than interchangeable under the proposed spectral-group representation.

When architectural modifications are introduced, both Group 5 (without the spectral stem) and Group 6 (with separate encoders) exhibit moderate but consistent OA degradation compared with the full model. This suggests that both the initial projection and the shared backbone are important for coordinating the dual objectives and stabilizing representation learning. Altogether, Group 7 achieves the best or near-best OA across all datasets, validating the joint contribution of

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TABLE X
COMPONENT ABLATION RESULTS UNDER THE LINEAR EVALUATION PROTOCOL

Group	Variant	Mask type	Branch		Stem	Shared Encoder	Overall Accuracy (%)			
			Rec.	Con.			Berlin	Salinas	WHU	Houston
1	MAE (spatial)	Spatial	✓	×	✓	—	83.18 ± 0.17	71.41 ± 0.80	95.77 ± 0.18	59.15 ± 0.63
2	MAE (spectral)	Band-wise	✓	×	✓	—	75.49 ± 0.46	62.94 ± 1.66	90.55 ± 0.21	50.88 ± 1.18
3	MSG (rec only)	Spectral-group	✓	×	✓	—	84.51 ± 0.10	90.39 ± 0.55	**99.32 ± 0.07**	69.44 ± 1.57
4	MSG (con only)	Spectral-group	×	✓	✓	—	85.72 ± 0.07	92.75 ± 0.73	98.95 ± 0.15	80.32 ± 0.48
5	MSG w/o stem	Spectral-group	✓	✓	×	✓	85.86 ± 0.34	92.31 ± 0.89	99.21 ± 0.13	84.19 ± 0.61
6	MSG (sep. enc.)	Spectral-group	✓	✓	✓	×	86.32 ± 0.30	92.98 ± 1.03	99.14 ± 0.27	85.42 ± 0.53
7	MSG (full)	Spectral-group	✓	✓	✓	✓	**86.56 ± 0.21**	**93.41 ± 1.14**	99.27 ± 0.38	**86.30 ± 0.77**

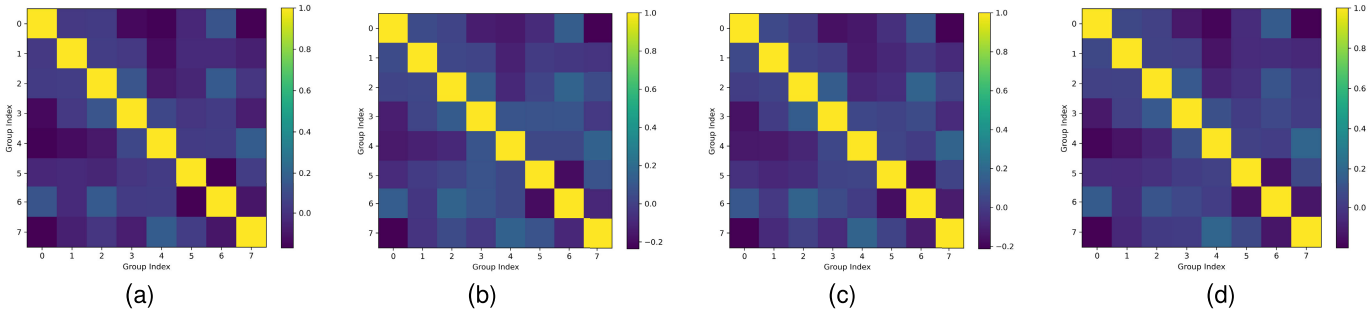


Fig. 3. Visualization of group-wise cosine similarity matrices on four downstream datasets. Each matrix is computed from a randomly sampled patch using the pretrained SGEM. Brighter diagonal entries and low off-diagonal responses indicate strong intragroup coherence and weak intergroup correlation. (a) Berlin. (b) Salinas. (c) WHU-Hi-LongKou. (d) Houston2013.

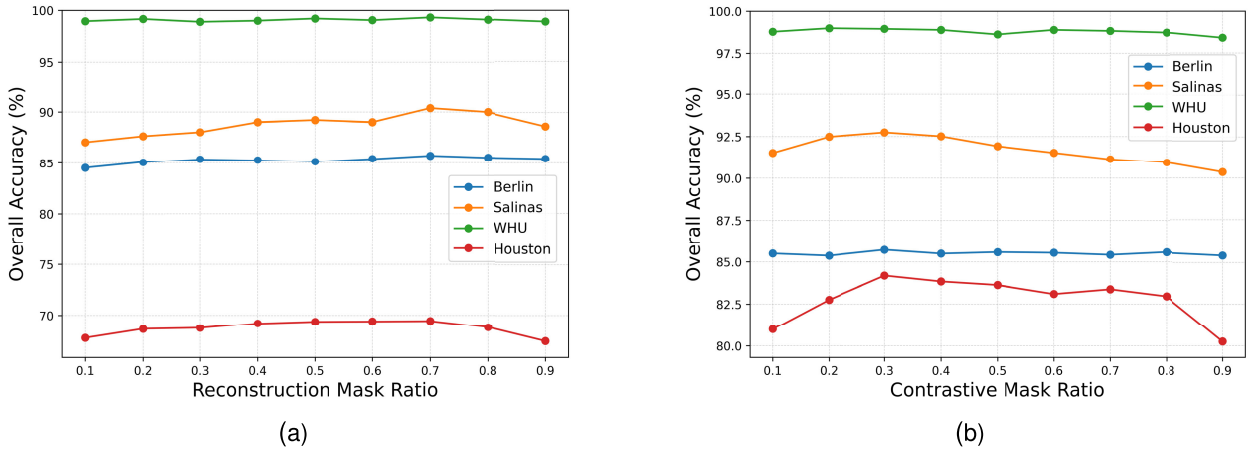


Fig. 4. Sensitivity to mask ratios for (a) reconstruction branch and (b) contrastive branch. Results are obtained under single-branch settings following Groups 3 and 4 of the component ablation in Table X.

spectral-group masking, coordinated dual-objective pretraining, and the shared encoder design.

2) Group Similarity Visualization: To better understand how the proposed spectral-group embedding facilitates redundancy reduction and promotes decorrelated spectral representations, we provide a qualitative visualization of group-wise similarities produced by SGEM. Specifically, for a randomly sampled patch, we compute a group prototype by averaging all spatial tokens within each spectral group, and then calculate the pairwise cosine similarity between group prototypes, resulting in a $G \times G$ similarity matrix.

Fig. 3 presents representative similarity matrices on four downstream datasets. Across all datasets, the matrices exhibit clear diagonal dominance, indicating strong intragroup coherence, while most off-diagonal entries remain close to zero, suggesting weak intergroup correlation. This pattern implies that different spectral groups capture complementary and largely nonredundant information, validating the ability of SGEM to organize spectral bands into coherent yet distinct groups.

Moreover, the highly consistent visualization patterns across datasets demonstrate that the learned grouping behavior is stable and dataset-agnostic, reflecting an intrinsic property of the proposed representation rather than a dataset-specific artifact. These qualitative observations provide intuitive evidence that group-wise embedding and masking encourage

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TABLE XI
EFFICIENCY AND COMPLEXITY COMPARISON

Method	Params (M)	FLOPs (M)	GPU Mem (MB)	Thr. (sps)
SpatialMAE	1.42	27.67	26.65	16396.03
SimCLR	0.88	79.46	47.44	14322.91
SS-MAE	2.83	99.73	56.94	8093.49
HSIMAE	1.80	351.34	222.37	5796.17
TMAC	2.73	86.70	55.10	7247.32
FactoFormer	1.03	67.85	44.84	3467.26
MSG	5.60	585.87	294.51	2542.84

TABLE XII
SENSITIVITY TO SPECTRAL GROUP SIZE g UNDER LINEAR EVALUATION

Metric	g=8	$\boldsymbol{g}$**=16**	g=32	g=64
Berlin OA	86.93 ± 0.21	86.56 ± 0.21	86.48 ± 0.14	86.38 ± 0.50
Berlin AA	81.05 ± 1.68	79.92 ± 1.00	82.19 ± 0.61	82.94 ± 0.61
Berlin κ	79.52 ± 0.40	78.80 ± 0.43	78.93 ± 0.35	78.92 ± 0.80
Salinas OA	93.86 ± 0.53	93.41 ± 1.14	93.17 ± 0.88	93.20 ± 1.03
Salinas AA	96.91 ± 0.33	96.51 ± 0.34	94.83 ± 0.44	95.78 ± 1.41
Salinas κ	93.20 ± 0.59	92.67 ± 1.26	92.40 ± 1.01	92.45 ± 1.22
WHU OA	99.15 ± 0.00	99.27 ± 0.38	99.03 ± 0.19	99.20 ± 0.15
WHU AA	99.31 ± 0.34	99.22 ± 0.30	98.55 ± 0.20	99.23 ± 0.39
WHU κ	98.85 ± 0.22	99.05 ± 0.50	98.73 ± 0.29	99.04 ± 0.24
Houston OA	88.61 ± 0.70	86.30 ± 0.77	86.12 ± 0.51	85.40 ± 0.74
Houston AA	90.59 ± 0.60	88.10 ± 0.97	88.09 ± 0.47	87.80 ± 0.63
Houston κ	87.71 ± 0.83	85.18 ± 0.84	85.07 ± 0.50	84.23 ± 0.85
FLOPs (M)	1184.86	585.87	292.67	147.64
Params (M)	18.19	5.60	2.45	1.67

TABLE XIII
SENSITIVITY TO ENCODER DEPTH L_{enc} UNDER LINEAR EVALUATION

Metric	L=2	L=4	$\boldsymbol{L}$**=6**	L=8
Berlin OA	86.15 ± 0.65	86.47 ± 0.14	86.56 ± 0.21	86.81 ± 0.35
Berlin AA	78.87 ± 2.32	80.56 ± 2.36	79.92 ± 1.00	82.86 ± 0.98
Berlin κ	78.14 ± 0.91	78.83 ± 0.49	78.80 ± 0.43	79.59 ± 0.51
Salinas OA	93.15 ± 0.58	91.33 ± 2.56	93.41 ± 1.14	93.91 ± 0.73
Salinas AA	96.03 ± 0.84	94.39 ± 1.83	96.51 ± 0.34	96.80 ± 0.44
Salinas κ	92.39 ± 0.67	90.35 ± 2.68	92.67 ± 1.26	93.18 ± 1.02
WHU OA	98.86 ± 0.29	98.86 ± 0.25	99.27 ± 0.38	99.06 ± 0.00
WHU AA	98.29 ± 0.63	98.57 ± 0.66	99.22 ± 0.30	99.10 ± 0.44
WHU κ	98.51 ± 0.25	98.55 ± 0.12	99.05 ± 0.50	98.70 ± 0.03
Houston OA	81.18 ± 0.48	84.04 ± 0.70	86.30 ± 0.77	89.08 ± 0.67
Houston AA	81.16 ± 0.55	85.45 ± 1.08	88.10 ± 0.97	91.11 ± 0.81
Houston κ	79.61 ± 0.66	82.74 ± 0.83	85.18 ± 0.84	88.18 ± 0.86
FLOPs (M)	239.25	412.56	585.87	759.18
Params (M)	4.81	5.21	5.60	6.00

TABLE XIV
SENSITIVITY TO EMBEDDING DIMENSION d UNDER LINEAR EVALUATION

Metric	d=64	$\boldsymbol{d}$**=128**	d=192	d=256
Berlin OA	86.08 ± 0.29	86.56 ± 0.21	87.09 ± 0.42	87.67 ± 0.29
Berlin AA	81.81 ± 0.73	79.92 ± 1.00	82.29 ± 2.36	84.55 ± 0.74
Berlin κ	78.47 ± 0.44	78.80 ± 0.43	79.84 ± 0.68	80.96 ± 0.71
Salinas OA	92.69 ± 1.14	93.41 ± 1.14	93.66 ± 0.34	93.85 ± 0.84
Salinas AA	95.23 ± 1.35	96.51 ± 0.34	96.91 ± 0.26	96.85 ± 0.12
Salinas κ	91.85 ± 1.18	92.67 ± 1.26	92.90 ± 0.31	93.21 ± 1.07
WHU OA	99.12 ± 0.00	99.27 ± 0.38	99.13 ± 0.02	99.10 ± 0.09
WHU AA	98.36 ± 0.02	99.22 ± 0.30	99.27 ± 0.32	99.27 ± 0.30
WHU κ	98.85 ± 0.21	99.05 ± 0.50	98.83 ± 0.06	98.89 ± 0.14
Houston OA	84.82 ± 0.68	86.30 ± 0.77	87.75 ± 0.74	89.28 ± 0.64
Houston AA	86.98 ± 0.73	88.10 ± 0.97	90.00 ± 0.71	91.26 ± 0.42
Houston κ	83.62 ± 0.92	85.18 ± 0.84	86.78 ± 0.91	88.49 ± 0.78
FLOPs (M)	189.03	585.87	1244.85	2165.98
Params (M)	1.54	5.60	12.35	21.78

TABLE XV
SENSITIVITY TO THE BALANCING WEIGHT λ UNDER LINEAR EVALUATION

Metric	λ=0.2	λ=0.33	$\boldsymbol{\lambda}$**=0.5**	λ=0.67	λ=0.8
Berlin OA	86.21 ± 0.51	86.35 ± 0.23	86.56 ± 0.21	86.49 ± 0.40	86.35 ± 0.45
Berlin AA	78.72 ± 1.34	79.30 ± 1.07	79.92 ± 1.00	80.10 ± 1.00	80.37 ± 1.01
Berlin κ	78.18 ± 0.78	78.45 ± 0.43	78.80 ± 0.43	78.73 ± 0.69	78.54 ± 0.53
Salinas OA	93.06 ± 1.01	93.18 ± 0.47	93.41 ± 1.14	93.18 ± 0.80	92.94 ± 1.34
Salinas AA	96.31 ± 0.48	96.43 ± 0.22	96.51 ± 0.34	96.40 ± 0.22	96.42 ± 0.38
Salinas κ	92.14 ± 1.48	92.40 ± 0.52	92.67 ± 1.26	92.41 ± 0.88	92.28 ± 1.12
WHU OA	99.08 ± 0.29	99.24 ± 0.29	99.27 ± 0.38	99.27 ± 0.13	99.18 ± 0.27
WHU AA	98.47 ± 0.47	98.53 ± 0.34	99.22 ± 0.30	98.10 ± 0.16	98.27 ± 0.57
WHU κ	98.80 ± 0.38	99.01 ± 0.38	99.05 ± 0.50	99.05 ± 0.16	98.92 ± 0.36
Houston OA	86.23 ± 0.87	86.27 ± 0.81	86.30 ± 0.77	85.60 ± 0.44	84.79 ± 0.60
Houston AA	88.11 ± 0.53	88.07 ± 0.98	88.10 ± 0.97	87.21 ± 0.32	86.60 ± 0.47
Houston κ	85.11 ± 0.94	85.14 ± 0.88	85.18 ± 0.84	84.43 ± 0.47	83.55 ± 0.64

decorrelated spectral representations, which complements the quantitative ablation results in Section IV-D1 and helps explain the performance gains of MSG.

3) Efficiency and Complexity Analysis: To complement accuracy-oriented comparisons, we analyze the computational complexity and practical inference efficiency of MSG and representative baselines. We report parameters, FLOPs, peak GPU memory, and throughput under a unified patch-level setting. All methods take hyperspectral patches of shape $[B, 128, 7, 7]$. Params and FLOPs are computed for a single forward pass with $B = 1$ (per-sample complexity). Peak GPU memory is measured by `torch.cuda.max_memory_allocated` (MB), and throughput is measured as samples per second (sps); both are obtained with $B = 64$ under forward-only inference on the same GPU.

Table XI summarizes the results. MSG incurs the highest computation and memory cost among the compared methods, leading to a lower forward throughput under the same batch size. This overhead is expected because MSG introduces spectral-group modeling and couples two complementary supervision signals (masked reconstruction and contrastive discrimination), which increases intermediate activations and computation relative to single-objective designs.

Despite being more demanding, MSG remains practical in patch-based hyperspectral pipelines: the peak allocated memory stays below a few hundred MB for 7×7 patches, and the measured throughput supports both pretraining and downstream FT. Combined with the consistent performance gains reported in Section IV, these results indicate an accuracy–efficiency tradeoff: MSG is preferable when stronger transferability/robustness is required, whereas lighter baselines may be chosen when computational resources are the primary constraint.

4) Summary of Hyperparameter Sensitivity: We further perform extensive sensitivity studies on key hyperparameters,

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TABLE XVI
SENSITIVITY TO PRETRAINING SCALE N UNDER LINEAR EVALUATION

Metric	N=50k	$\boldsymbol{N}$**=100k**	N=200k	N=400k
Berlin OA	86.20 ± 0.16	86.56 ± 0.21	86.56 ± 0.29	86.63 ± 0.28
Berlin AA	79.34 ± 1.45	79.92 ± 1.00	80.37 ± 0.89	80.44 ± 1.06
Berlin κ	78.24 ± 0.16	78.80 ± 0.43	78.84 ± 0.30	78.95 ± 0.52
Salinas OA	92.71 ± 0.29	93.41 ± 1.14	93.18 ± 0.88	93.65 ± 1.14
Salinas AA	95.98 ± 0.56	96.51 ± 0.34	96.28 ± 0.21	96.52 ± 0.36
Salinas κ	91.88 ± 0.32	92.67 ± 1.26	92.40 ± 0.97	92.93 ± 1.26
WHU OA	98.80 ± 0.26	99.27 ± 0.38	99.46 ± 0.16	99.49 ± 0.15
WHU AA	97.70 ± 0.46	99.22 ± 0.30	98.75 ± 0.24	98.62 ± 0.17
WHU κ	98.43 ± 0.34	99.05 ± 0.50	99.30 ± 0.21	99.34 ± 0.20
Houston OA	85.46 ± 0.69	86.30 ± 0.77	87.11 ± 0.87	87.78 ± 0.58
Houston AA	87.40 ± 0.57	88.10 ± 0.97	89.13 ± 0.53	89.38 ± 0.64
Houston κ	84.27 ± 0.75	85.18 ± 0.84	86.06 ± 0.94	86.78 ± 0.63

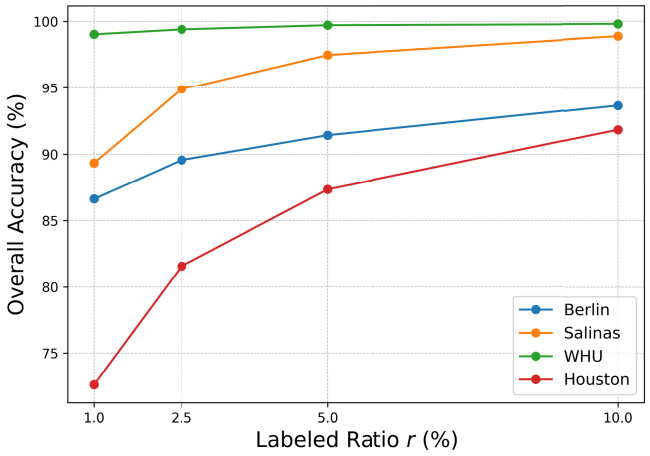


Fig. 5. OA versus labeled ratio r under FT.

TABLE XVII
OA COMPARISON BETWEEN FS (SCRATCH) AND FT (PRETRAINED)

Method	Berlin	Salinas	WHU	Houston
FS (scratch)	78.17 ± 1.01	82.21 ± 1.65	94.62 ± 0.87	75.07 ± 2.49
FT (pretrained)	83.68 ± 0.53	86.95 ± 1.86	97.78 ± 0.26	87.35 ± 0.34

including mask ratios for the reconstruction and contrastive branches, spectral group size, encoder depth, embedding dimension, and other training settings. Overall, MSG remains stable within a reasonably wide range and achieves a favorable performance–efficiency balance under the default configuration. It is worth noting that linear evaluation may underestimate the impact of model capacity, since the shallow classifier constrains the expressiveness of learned representations. Nevertheless, consistent performance–efficiency trends can still be observed, motivating our choice of a balanced configuration. To keep the main text concise, detailed hyperparameter tables and additional analyses are deferred to Appendix A.

V. CONCLUSION

In this article, we present MSG, a joint MSG framework for hyperspectral representation learning. Rather than loosely combining masked reconstruction and contrastive learning, MSG formulates an objective-consistent joint pretraining mechanism that couples the two objectives through a shared spectral-group representation and coordinated group-wise masking. By organizing contiguous bands into coherent groups and enforcing consistent grouped views under complementary masking ratios, MSG suppresses redundancy-induced shortcut solutions while preserving informative spectral context, so that reconstruction-driven fidelity learning and contrastive-driven discrimination can complement rather than interfere with each other. Experiments under linear evaluation and few-shot FT on four downstream benchmarks that span diverse scenes and sensors demonstrate that MSG consistently outperforms strong hyperspectral self-supervised baselines and vision-transfer counterparts. Comprehensive ablation studies and supplementary analyses further verify the contribution of each key component and provide practical guidance on hyperparameter robustness and efficiency tradeoffs.

Despite these improvements, several limitations remain. First, our current evaluation mainly focuses on hyperspectral classification, which may not fully reflect the generality of the learned representations. Second, while spectral grouping improves robustness and coordination, it introduces additional computational overhead, and a more efficient implementation is desirable for large-scale or high-resolution settings. Future work will extend MSG to broader downstream tasks such as segmentation, change detection, and anomaly detection, and explore scaling MSG to larger pretraining corpora and cross-sensor or multimodal pretraining, with further optimization of efficiency and training stability.

APPENDIX

This appendix provides supplementary results supporting Section IV-D, mainly focusing on detailed hyperparameter sensitivity under linear evaluation and additional FT analyses. Unless otherwise stated, results are reported as mean±std over multiple seeds. For conciseness in tables and figures, we abbreviate WHU-Hi-LongKou as "WHU" and Houston2013 as "Houston."

A. Detailed Hyperparameter Sensitivity

This section reports detailed sensitivity results of key hyperparameters under the linear evaluation protocol. We highlight the adopted default configuration by boldfacing the corresponding column header.

1) Mask Ratio: To isolate the effects of the two pretraining objectives, we vary the mask ratios under single-branch configurations, where only one branch is active at a time. Specifically, the reconstruction sweep activates the reconstruction branch, whereas the contrastive sweep activates the contrastive branch. As shown in Fig. 4, performance peaks at $\tau_h = 0.7$ for reconstruction and $\tau_l = 0.3$ for contrastive learning across datasets. A higher reconstruction masking ratio encourages the model to infer missing spectral–spatial context, promoting holistic feature learning. Conversely, a lower contrastive masking ratio preserves sufficient anchor information for stable view alignment and robust discrimination. These ratios are adopted as the default setting throughout the article.

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2) Spectral Group Size: Table XII reports the sensitivity to the spectral group size $g \in \{8, 16, 32, 64\}$, which controls the granularity of spectral tokenization. Smaller g yields finer-grained tokens and tends to improve accuracy but increases computational cost (FLOPs/Params). Considering the performance–efficiency tradeoff, we adopt $g = 16$ as the default.

3) Encoder Depth: Table XIII evaluates encoder depth $L \in \{2, 4, 6, 8\}$. Increasing depth typically improves representation capacity and benefits performance, while the computational cost grows roughly linearly with L. We adopt $L = 6$ as a balanced default setting.

4) Encoder Dimension: Table XIV studies the embedding dimension $d \in \{64, 128, 192, 256\}$. Larger d generally increases capacity and improves accuracy, but FLOPs/Params grow rapidly with d. We adopt $d = 128$ as the default for a favorable performance–efficiency tradeoff.

5) Balancing Weight: We further examine the sensitivity to the balancing weight λ, which controls the tradeoff between masked reconstruction and contrastive discrimination during joint pretraining. Table XV shows that MSG remains stable within a moderate range around the default $\lambda = 0.5$. This behavior suggests that the two objectives are well-coupled by the shared spectral-group representation and do not require delicate reweighting to work effectively.

6) Pretraining Scale: Finally, we study the impact of pretraining scale by varying the number of pretraining iterations/samples $N \in \{50\text{k}, 100\text{k}, 200\text{k}, 400\text{k}\}$, while keeping all other settings fixed. As shown in Table XVI, increasing N generally improves performance and stabilizes representation learning, with diminishing returns at larger scales. We adopt $N = 100\text{k}$ as the default to balance accuracy and computational cost. Notably, the consistent gains across datasets indicate that MSG benefits from additional unlabeled data without changing the training recipe, which supports its scalability. This also suggests that the framework can be readily extended to larger corpora when computational resources allow.

B. Additional FT Analyses

This section provides supplementary FT results that complement the main experiments in Section IV-D, focusing on robustness under limited labeled supervision and a comparison to training from scratch (FS).

1) Robustness to Limited Labels in FT: To evaluate robustness under limited supervision, we vary the labeled ratio r in FT using a synchronized split protocol: r for training, r for validation, and $1 - 2r$ for testing. We consider $r \in \{1\%, 2.5\%, 5\%, 10\%\}$ and report results as mean $\pm$ std over multiple random seeds. In this ablation, stride-2 is used for all datasets to ensure sufficient training samples, especially for minority classes, at very small labeled ratios such as $r = 1\%$. Therefore, these results are not directly comparable to the main FT results, where the default stride is 7 for Berlin, Salinas, and WHU, and 2 for Houston.

As shown in Fig. 5, performance improves consistently as more labeled samples become available. MSG remains competitive even at very small r, indicating that the pretrained representation transfers reliably to few-shot FT. Due to the stride-2 sampling used in this ablation, the accuracies for Berlin, Salinas, and WHU at $r = 5\%$ are generally higher than those reported in the main text, while the Houston results remain consistent.

2) Comparison to Training FS: We further compare fully supervised training FS with FT from pretrained weights. Both settings follow the same 5/5/90 split (5% training, 5% validation, and 90% testing). For a fair comparison, the architecture and optimization settings are kept identical, and only the initialization differs.

As shown in Table XVII, FT consistently outperforms FS across all four datasets under the same split. The gains are particularly pronounced in Houston, where FT improves OA by more than 12 percentage points, while clear improvements are also observed in Berlin, Salinas, and WHU. These results indicate that the proposed pretraining learns transferable spatial–spectral representations that provide a substantially better initialization than random weights. In addition to improving the final accuracy, pretrained initialization also leads to lower variance on most datasets, suggesting more stable optimization in the low-label regime.

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